Data Science and AI for Finance
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Session 4 – Financial Products and Pricing

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TOC of financial environment

- Financial environments generally fall into four distinct "Architectures":
  - Centralized Exchange (CLOB): Many-to-Many matching (High transparency).
  - Dealer Networks (OTC): Hub-and-Spoke (Negotiated, lower transparency).
  - Commercial and Retail Bilateral (Direct Origination): One-to-One (Bilateral contracts).
  - Retail Collective (Risk Pooling-Actuarial): Many-to-One (Transfer of liability).

Architecture 1: Centralized Exchange (CLOB)

- The "Public Square" – Many-to-Many Matching

- **Core Concept:** A centralized order book where all buy and sell orders are aggregated in a single venue. Matching is deterministic based on Price/Time priority.
- **Primary Examples:**
  - **Equities:** NYSE, NASDAQ, Tokyo Stock Exchange.
  - **Derivatives:** CME Group (Futures), CBOE (Options).
  - **Crypto:** Binance, Coinbase Pro (Spot markets).
- **Typical Products:**
  - Standardized Equities (Common Stock).
  - ETFs & Index Funds.
  - Standardized Futures & Options contracts.
- **Market Size (Est.):**
  - Global Equity Market Cap: ~$110 Trillion.
  - High frequency of turnover (Daily volume in Trillions).

Architecture 1: Centralized Exchange (CLOB)

- **Pricing Mechanism: Deterministic & Auction-Based.**
  - Price is discovered where the highest Bid meets the lowest Ask.
  - Continuous double-auction during trading hours.
- **Transparency: High (Pre- & Post-Trade).**
  - Order Book Visibility: Market depth (L2/L3 data) is visible to all participants.
  - Trade Reporting: Real-time dissemination of execution price and volume.
- **Liquidity: High / Centralized.**
  - Instant execution for small-to-medium size.
  - Liquidity is provided by HFTs, Market Makers, and Limit Orders from diverse participants.
- **Customization: None (Standardized).**
  - Assets must be fungible. You cannot negotiate contract terms (e.g., strike price or settlement date) individually.

Architecture 2: Dealer Networks (OTC)

- The "Bazaar" – Hub-and-Spoke Negotiation

•**Core Concept:** A decentralized network of "Dealers" (Banks/Desks) who hold inventory and quote buy/sell prices to clients. Clients trade *against* the dealer, not each other.
•**Primary Examples:**
•**Fixed Income:** US Treasuries, Corporate Bonds, Municipal Bonds.
•**Currencies:** Global Forex (FX) Spot & Forwards.
•**Derivatives:** Interest Rate Swaps (IRS), CDS (Credit Default Swaps).
•**Typical Products:**
•Bonds (Government & Corporate).
•Currency pairs.
•Complex structured products.
•**Market Size (Est.):**
•Global Bond Market: ~$130 Trillion.
•Global FX Daily Volume: ~$7.5 Trillion.

Architecture 2: Dealer Networks (OTC)

- **Pricing Mechanism: Probabilistic / Negotiated.**
  - Quote-driven: "Request for Quote" (RFQ). Prices vary by dealer and time (probabilistic).
  - Includes a "Spread" (Bid-Ask) which acts as the dealer's profit margin.
- **Transparency: Low to Medium.**
  - Pre-trade: Opaque (You only see the price quoted to you).
  - Post-trade: Delayed (TRACE reporting exists for bonds, but lags CLOB speed).
- **Liquidity: Fragmented.**
  - Depends on the dealer's balance sheet and risk appetite.
  - Can evaporate instantly during stress events (liquidity crunch).
- **Customization: Medium to High.**
  - Dealers can structure specific maturity dates or notional amounts to fit client needs.

Architecture 3: Direct Origination (Retail/Private)

- **Core Concept:** The *creation* of a new financial asset through a direct agreement between a capital provider and a user. Unlike Exchange or Dealer architectures (which trade *existing* assets), this architecture brings the asset into existence.
- **The Two Sub-Sectors:**
  - **Mass Retail (The "Menu"):** High-volume, standardized contracts for individual consumers. The goal is efficiency and scale.
  - **Private/Commercial (The "Bespoke"):** Low-volume, highly customized contracts for businesses. The goal is specific risk structuring.
- **Primary Examples:**
  - *Retail:* Residential Mortgages, Auto Loans, Credit Cards.
  - *Commercial:* Private Equity Buyouts, Venture Debt, Direct Lending, Project Finance.
- **Market Size (Est.):**
  - **Dominant:** The largest pool of capital globally (Real Estate Debt > $100 Trillion).
  - **Growth:** Private Credit (~$1.7 Trillion) is rapidly taking share from Dealer Networks (syndicated loans that are sliced up and traded OTC).

Architecture 3: Direct Origination (Retail/Private)

- **Pricing Mechanism: Bifurcated (Scorecard vs. Model).**
  - *Retail:* **Table-Driven.** Pricing is determined by a rigid matrix (Credit Score + Cost of Funds). It is deterministic for the applicant ("Take it or leave it").
  - *Commercial:* **Negotiated.** Pricing is derived from deep due diligence (Cashflow Models) and bilateral bargaining.
- **Customization: The Spectrum.**
  - *Retail:* **None.** Borrowers select from a fixed menu of templates (e.g., "15-year fixed vs. 30-year ARM").
  - *Commercial:* **Maximum.** Every clause is negotiable (Covenants, Amortization schedules, Equity Warrants, Board Seats).
- **Transparency: Private (Siloed).**
  - There is no "Ticker," "Order Book," or "Last Sale" feed.
  - Information exists *only* between the two parties (Lender & Borrower).
  - *Retail Privacy:* Your specific rate and terms are private data, even if the product is generic.
- **Liquidity: Illiquid (Held-to-Maturity).**
  - The asset is usually held on the originator's balance sheet for its lifespan.
  - *The "Exit" Exception:* Retail loans can be bundled into securities (MBS/ABS) to be sold in **Architecture 2** (Dealer Networks), but the original loan remains a bilateral contract.

Architecture 4: Retail (collective) Actuarial

- **Core Concept:** Participants transfer "Idiosyncratic Risk" (individual chance of loss) to a central entity (Pool) in exchange for a premium. The Pool relies on the Law of Large Numbers.

- **Primary Examples:**
  - **Insurance:** P&C (Property & Casualty), Life Insurance, Health.
  - **Reinsurance:** Munich Re, Swiss Re (Insuring the insurers).
  - **Govt Programs:** Social Security, FDIC Deposit Insurance.
- **Typical Products:**
  - Auto/Home Policies.
  - Annuities (Longevity risk pooling).
  - Catastrophe Bonds (Cat Bonds).
- **Market Size (Est.):**
  - Global Insurance Premiums: ~$7 Trillion annually.
  - Total Assets Under Management (Insurers): ~$40+ Trillion.

Architecture 4: Retail (collective) Actuarial

- **Pricing Mechanism: Actuarial / Statistical.**
  - Price is based on "Expected Loss" of the pool, not the individual asset.
  - Driven by data science, historical probabilities, and correlation models.
- **Transparency: Low (Black Box).**
  - The pricing model is proprietary intellectual property of the insurer.
  - The insured rarely understands the "True Cost" of the risk.
- **Liquidity: Contingent.**
  - Liquidity is only accessed upon a "Trigger Event" (Claim).
  - You cannot "sell" your car insurance policy to a third party.
- **Customization: Menu-Based (Modular).**
  - Standardized policies with optional "Riders" (add-ons).
  - Not fully bespoke, but tailored to risk profiles.

Overview of Product pricing

1. CLOB
    - Common stock overview
    - Common stock (case studies)
    - Indices and ETFs
    - Futures
    - Options
2. OTC
    - Bond Pricing
3. Retail (bilateral)
    - Delinquency Modeling (case in the credit section)
    - Interest on Loans and EMIs
4. Retail (Insurance)
    - Insurance pricing (case study)
    - Annuities (case study in personal finance section)

Stock pricing

- What determines the price of a stock?

Macro and Contextual factors

- **Global economy**: Inflation, interest rates, GDP growth, monetary policy.
- **Politics & regulation:** Tax laws, trade policies, political stability, lobbying power.
- **Industry competition:** Market share battles, disruptive entrants, mergers & acquisitions.
- **Technology & patents:** Innovation pipelines, intellectual property protection, R&D breakthroughs.
- **Social trends**: Consumer preferences, ESG (environmental/social/governance) pressures, demographic shifts.

Valuation & Fundamentals

- **Discounted Cash Flow (DCF):** Value of expected future cash flows, discounted back to present.
- **Dividend Discount Model (DDM):** Value of future dividends, discounted at required rate of return.
- **Relative valuation:** Price-to-earnings (P/E), price-to-book (P/B), EV/EBITDA compared to peers.
- **Balance sheet strength:** Debt levels, liquidity, asset quality.
- **Earnings growth:** Analyst forecasts, quarterly results, guidance.

Market Microstructure

- **Investor psychology:** Fear vs greed, momentum chasing, herd behavior.
- **Institutional flows:** Hedge funds, mutual funds, pension funds moving large volumes.
- **Liquidity:** How easily shares can be bought/sold without moving the price.
- **News catalysts:** Earnings surprises, lawsuits, product launches, CEO changes.
- **Speculation:** Options activity, short squeezes, algorithmic trading.

Ultimate Driver! Supply and Demand of buyers and sellers.

a stock's price is not just "what it's worth" but "what buyers and sellers agree on right now," with every outer influence funneling into that negotiation.

Case studies

Stock Pricing

1. Temporal methods
   a. Signal/Feature based Regression(Naive Approach)
   b. Time Series Modeling
2. Cross Sectional Approaches
   a. Value vs Growth categorizan
   b. GNN based approach for identifying relationship across stocks.

Indices

- Market indices are statistical measures that track the performance of a group of securities.
- Why do they exist?
  - **The Barometer (Information):** They give a single number to represent the health of an economy or sector. If the S&P 500 is down 20%, the US economy is likely in trouble.
  - **The Benchmark (Accountability):** Active fund managers charge high fees to "beat the market." The index is the yardstick used to prove if they are actually generating *Alpha* or just charging you for *Beta*.
  - **The Product (Access):** With the rise of ETFs (Exchange Traded Funds), indices went from being "measurements" to "investments." You can now buy the index itself.
- Types
  - Of weighting: Market cap, price and equal weighted.
  - Of sectors: Broad market, sector specific, style (growth/value), geography
- Example of market cap vs price

Indices

- How are index values calculated:
  - Pick the stocks
  - Calculate market cap = share price x shares outstanding
  - Calculate free-float market cap by adjusting each stock with a float factor
  - Add across all stocks.
  - Adjust by base value

$$\text{Nifty 50 Index value} = \frac{\text{Current Total Free - Float Market Cap}}{\text{Base Market Cap}} \text{ x } 1000$$

  - Over time, the 1000/base Market Cap is kept as a divisor, which is adjusted for bonus issues, splits, dividends, rights, mergers, entry and exit of stocks from the index. The divisor is not static.

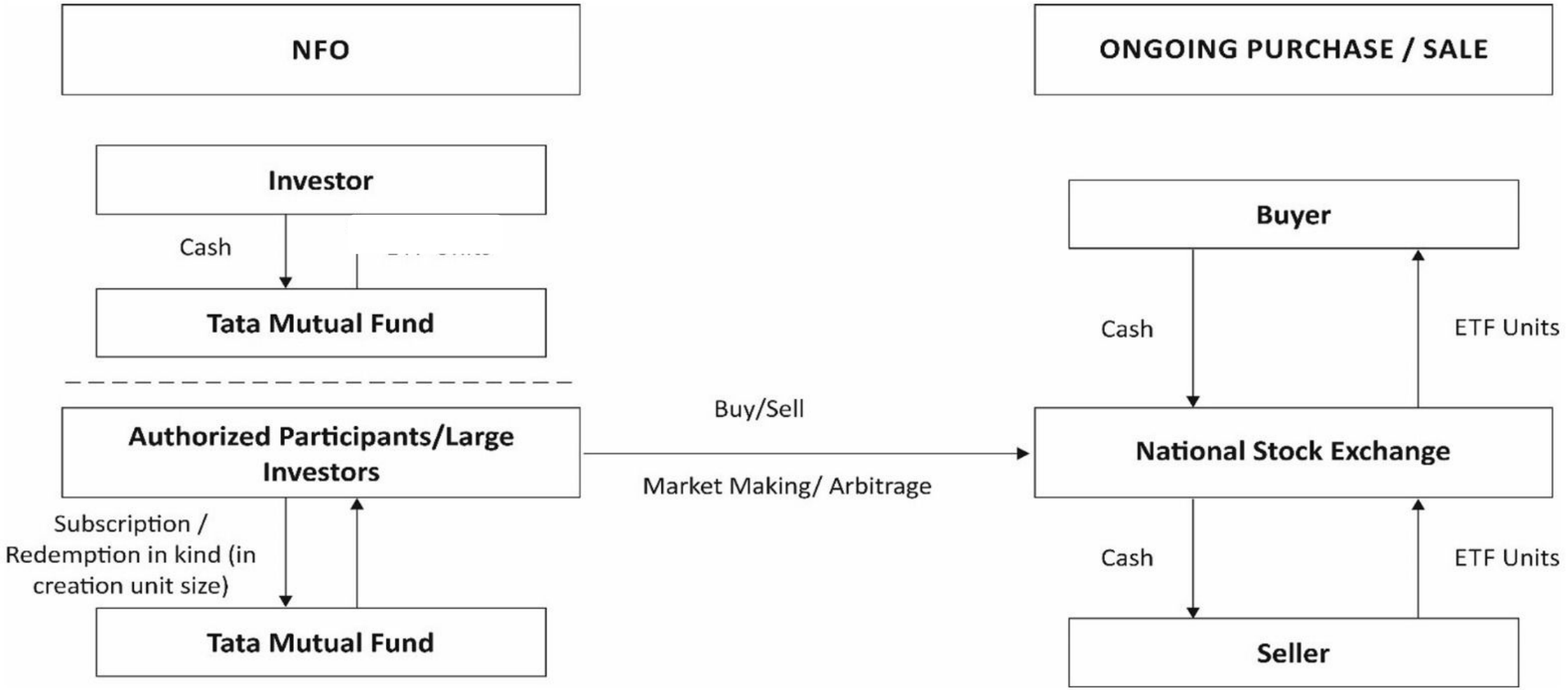
ETFs

- Definition: Exchange-Traded Funds (ETFs) are pooled investment vehicles that trade on stock exchanges like individual shares.
- Structure: Hold baskets of securities (stocks, bonds, commodities, etc.) designed to track an index or theme. So, how is it different from mutual funds?
- Features:
    - Liquidity: Bought/sold throughout the trading day.
    - Transparency: Holdings often disclosed daily.
    - Diversification: One trade gives exposure to many securities.
    - Cost efficiency: Lower fees than many mutual funds.
- Examples: Nifty 50 ETF, S&P 500 ETF, Gold ETF.

ETFs

- Authorised Participants (APs): Large institutions that interact directly with the ETF issuer.
- Creation process:
  1. AP delivers a basket of underlying securities (or cash) to the ETF issuer.
  2. In return, AP receives Creation Units (large blocks of ETF shares).
  3. AP can sell ETF shares on the exchange to investors.
- Redemption process:
  1. AP returns ETF shares to the issuer.
  2. Receives the underlying basket of securities (or cash) back.
- Why it matters:
  - Keeps ETF price close to Net Asset Value (NAV).
  - Provides liquidity and arbitrage opportunities.
  - Ensures efficient functioning of ETFs in the market.

How ETF's Works:



Futures

- What are futures: A future is a standardized financial contract where two parties agree to buy or sell an asset at a predetermined price on a specific future date.
- It is a derivative (what are some others?).
- What are the type of futures: Equity Index, Stock futures, commodity futures, currency futures, Interest Rate futures (bonds and libor), crypto futures.
- Cash settlement vs physical delivery
- Mark-to-market
- Leverage

Pricing Futures

- How do I price them?
  - The probability answer: Just like a stock, build PDF of possible future value of the stock at expiry. Take expected value, adjust for risk? Adjust for margin (loss of interest)?
  - Wrong Answer!
  - The price of a Future is determined by arbitrage, not by prediction.
  - How do we replicate the future with stock. If: S_0:stock price today, F_0: future price today, r is risk-free interest rate, T is the time till expiry, and $S_T$is stock/future price at expiry.
  - Then cash-and-carry would cost $S_T - S_0 e^{rT}$
  - The long future would cost $S_T - F_0$,
  - How are currency futures and commodity futures different from the $F_0 = S_0 e^{rT}$

Pricing futures

- Commodity futures have Exchange for Physical cost
- Currency futures account for interest rate of borrowing as well as the depositing interest rates so $F_0 = S_0 \frac{e^{r_{deposit}T}}{e^{r_{borrow}T}} = S_0 e^{(r_d - r_b)T}$
- Simple example:

•**Spot USD/INR:** $S_0 = 80$ INR per USD
•**Domestic (INR) interest rate:** $r_{INR} = 5\%$ per year
•**Foreign (USD) interest rate:** $r_{USD} = 2\%$ per year
•**Tenor:** T=1 year
•**Observed 1-year USD/INR futures price:** F_0=81 INR per USD
Step 1: Compute the theoretical forward price
Formula (oversimplified to illustrate the point): $F^* = S_0 \frac{(1+r_{INR})T}{(1+r_{USD})T}$
Plugging in: $80 \frac{(1.05)}{(1.02)} = 82.35$
So the **fair future** should be about 82.35, but the market is quoting only 81.

Arbitrage strategy (since $F_0 < F^*$)

1. **Borrow USD 100,000** at 2% for 1 year.
   •Repayment due: 100,000 * 1.02=102,000 USD.
2. **Convert USD to INR spot**:
   •Receive 100,000 * 80=8,000,000 INR.
3. **Invest INR at 5% for 1 year**:
   •Value at maturity: 8,000,000 * 1.05=8,400,000 INR.
4. **Enter futures contract to buy USD at 81**:
   •To repay 102,000 USD, you need INR 102,000 * 81 = 8,262,000.

5. **Pocket the difference** of 8,400,000 – 8,262,000 = 138,000 INR

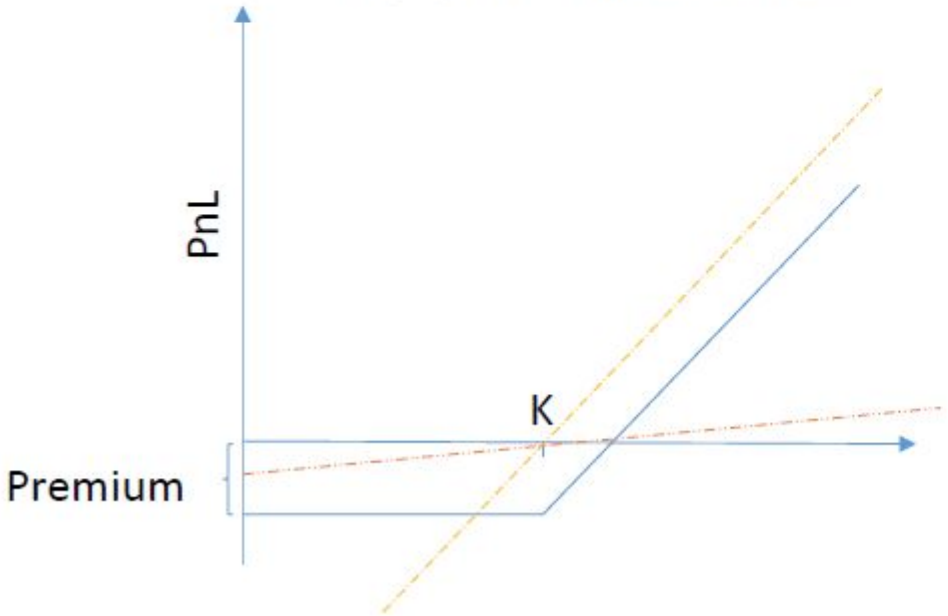
Options

- Definition: Financial contracts giving the right (not obligation) to buy or sell an asset at a set price before/at expiry.
- Underlying assets: Stocks, indices, currencies, commodities, futures, ETFs.
- Key terms:
  - Strike price (K): Agreed price for the transaction.
  - Expiration date (T): Last date the option can be exercised.
- Premium: Price paid to acquire the option.
- Two main types: Calls and Puts.
- American and European Options

Options

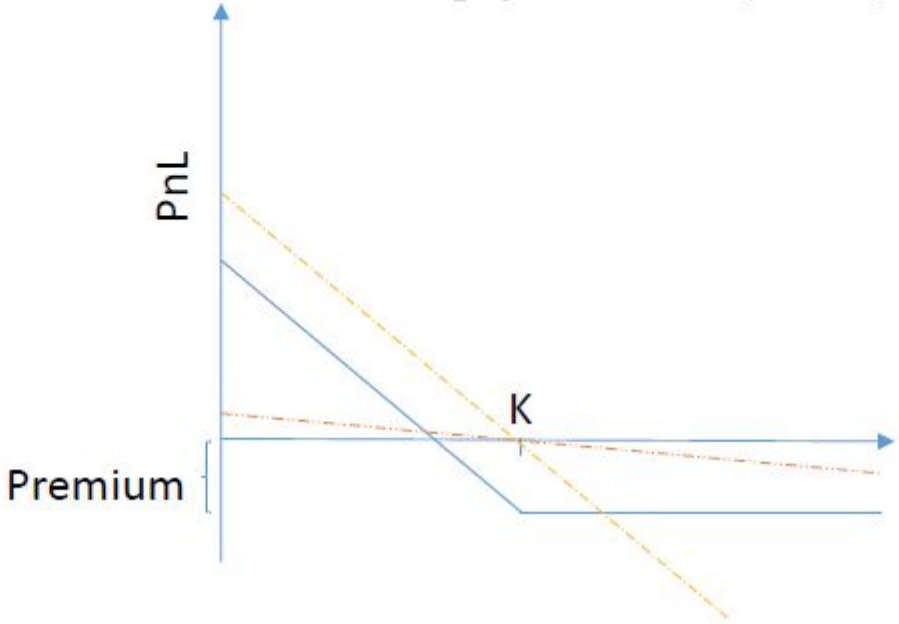
- What is a Call Option?
- Right to buy the underlying asset at strike price K.
- Buyer's perspective: Gains if asset price rises above strike.
- Payoff at expiry:

$$\textbf{Call payoff} = \max(S_T - K, 0)$$



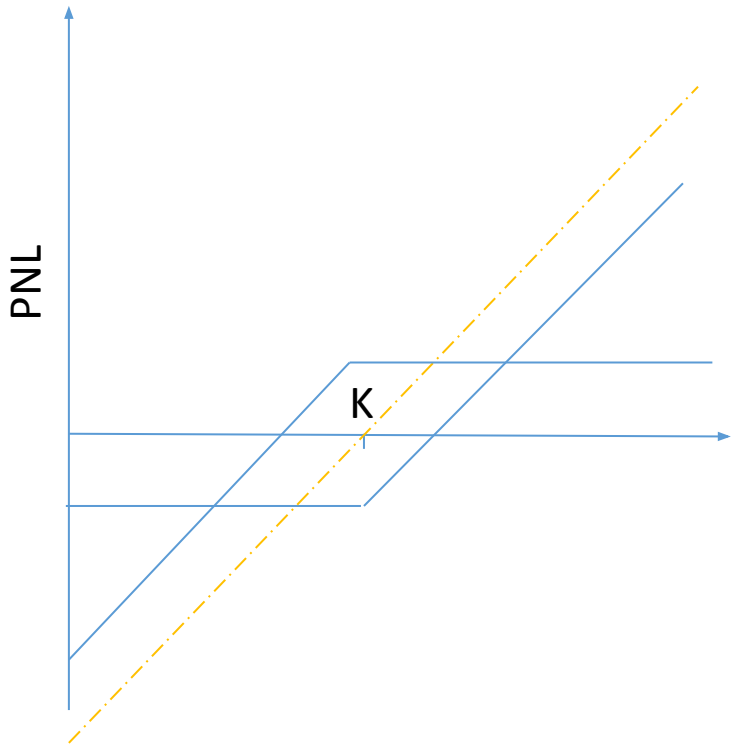
- What is a Put Option?
- Right to Sell the underlying asset at strike price K.
- Buyer's perspective: Gains if asset price drops below strike.
- Payoff at expiry:

$$\textbf{Put payoff} = \max(K - S_T, 0)$$



Put-Call Parity

- $C - P = S_0 - Ke^{-rT}$



- Ensures no arbitrage between calls, puts, and stock/future.
- Interpretation 1: Long Call + Short put is the same as buying the stock and borrowing the PV of the strike.
- Interpretation 2: A call + discounted strike cash invested = put + stock.
- Interpretation 3: Long Call + Short Put + discounted strike in cash is the same as purchasing the stock
- What is the payout of C-P? $S_T - K$
- What is payout of $S_0 - Ke^{-rT}$? $S_T - K$

Greeks

- Delta (Δ): Sensitivity of option price to changes in underlying.
- Gamma (Γ): Rate of change of Delta — captures convexity.
- Theta (Θ): Time decay — how option value erodes as expiry nears.
- Vega (v): Sensitivity to volatility changes.
- Rho (ρ): Sensitivity to interest rate changes.
- Purpose: Risk management, hedging, and trading strategies.

Bond Pricing

- For a bond with:

C = coupon payment per period
F = face value (principal)
r = discount rate (yield per period)
n = number of periods until maturity

$$P = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{F}{(1+r)^n}$$

- What happens between time periods? The saw tooth effect of coupon payments (dirty price vs clean price)
- The effect of interest rate on bond price. Price negotiation is entirely on "r".
- Amortization of Premium and Accretion of discount. The migration to face value.

Interest on Loans and Delinquency models

- $\text{Interest Rate} = \text{Base Rate} + \text{Risk Premium} + \text{OpEx} + \text{Profit Margin}$
- 1. The Base Rate (The "Cost of Goods")This is the "Floor." It represents what it costs the lender to "buy" the money they are lending to you. For Banks: This is often tied to the Central Bank rate (e.g., Repo Rate or SOFR in the US) or their internal cost of deposits. This is close to the risk-free rate. For Private Lenders: This is the return their investors (LPs) demand just for showing up (around 0-8%).
- 2. The Risk Premium (The "Spread")This is the variable part that changes based on who is borrowing. It prices the probability of default. We will look at 2-3 cases on detailed modeling of default probability.
- 3. Operating Expenses (OpEx)The "Administrative Load." The bank has to pay for the branch, the software, and the loan officer's salary. They typically add 0.25% – 0.50% to cover these overheads.
- 4. Profit Margin (The "Vig")Finally, the bank adds their target profit. This is usually determined by their Return on Equity (ROE) targets.

From loan interest to EMIs

- How do we translate loan interest to EMIs?
  - If we paid only the interest, then we would pay forever.
  - If we paid a fixed and equal portion of the principal in each repayment then the amount outstanding would decrease and therefore the interest on the amount outstanding would decrease making the repayment unequal.
  - One workaround is the flat rate method (consumer durables). It calculates total payable as P+(P x R x N) and divides it by the tenure. This is more than the reducing balance.
  - How do we calculate the reducing balance EMIs, but maintain a constant payment schedule?

From loan interest to EMIs

- Let:
- P = Principal loan amount
- r = Monthly interest rate (in decimal)
- n = Total number of months
- E = EMI (monthly payment)
- The EMI is paid at the **end of each month**, from month 1 to month n. Each payment is discounted back to present value:
- Value of 1st EMI = $\frac{E}{(1+r)^1}$
- Value of 2nd EMI = $\frac{E}{(1+r)^2}$
- Value of 3rd EMI = $\frac{E}{(1+r)^3}$

- This is a geometric series
- WKT
- $S_n = a + ab + ab^2 + ab^3 + \cdots + ab^{n-1}$
- $bS_n = ab + ab^2 + ab^3 + \cdots + ab^{n-1} + ab^n$
- $S_n - bS_n = a - ab^n$
- $S_n(1-b) = a(1-b^n)$
- $S_n = a \cdot \frac{1-b^n}{1-b}$

From interest to EMIs

- In our problem a $= \frac{E}{(1+r)}$ and b $= \frac{1}{(1+r)}$

So:

$$P = E \cdot \frac{\frac{1}{1+r} \cdot \left[1 - \left(\frac{1}{1+r}\right)^n\right]}{1 - \frac{1}{1+r}}$$

$$1 - \frac{1}{1+r} = \frac{(1+r) - 1}{1+r} = \frac{r}{1+r}$$

$$P = E \cdot \frac{1}{1+r} \cdot \left[\frac{1 - \frac{1}{(1+r)^n}}{\frac{r}{1+r}}\right]$$

$$P = \frac{E}{r} \cdot \left[1 - \frac{1}{(1+r)^n}\right]$$

$$P = \frac{E}{r} \cdot \left[\frac{(1+r)^n - 1}{(1+r)^n}\right]$$

$$E = P \cdot r \cdot \frac{(1+r)^n}{(1+r)^n - 1}$$